Movie Ticket Booking System

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01

AP(OOP) LAB - Spring 2026

Problem Statement

Problem Statement:
A system to manage movie seat booking efficiently while preventing double booking using structured state management.

Target User:

- Students learning OOP
- Beginner Java developers
- Users simulating booking systems

Architecture and Key Classes

Architecture

- Presentation Layer → Main class (user input/output)
- Business Layer → MovieService, BookingService
- Data Layer → Movie, Show, Seat

Key Classes

- Movie → stores movie details
- Show → manages seat grid
- Seat → maintains state
- SeatState → defines behavior

02

03

OOP Concepts Used

01

Abstraction

- Implemented using the SeatState interface
- Defines common operations like book() and getSymbol()
- Hides internal logic of seat behavior from other classes

02

Inheritance

- AvailableState and BookedState implement SeatState
- Both reuse the same structure but provide different behavior

03

Polymorphism

- Method call state.book(this) behaves differently based on state
- If seat is available → gets booked
- If already booked → shows error

04

Design Pattern : State Pattern

- Each seat has a state object (Available or Booked)
- Behavior changes when state changes
- No need for complex if-else conditions

Technical Highlight

Key Design Decision
Using State Pattern for seat booking
Logic

- Each seat has a state object
- Booking depends on current state
- No if-else conditions needed

Why Chosen

- Cleaner code
- Easy to extend (Reserved, VIP, etc.)
- Avoids complex conditional logic

EXPLORER

MOVIEBOOKINGSYSTEM
- .vscode
  - launch.json
- AvailableState.class
- AvailableState.java
- BookedState.class
- BookedState.java
- BookingService.class
- BookingService.java
- Main.class
- Main.java 1
- Movie.class
- Movie.java
- MovieService.class
- MovieService.java
- Seat.class
- Seat.java
- SeatState.class
- SeatState.java
- Show.class
- Show.java

OUTLINE
- BookingService
  - bookSeat(Show, i...

TIMELINE

JAVA PROJECTS
- MovieBooki...
  - BookingService
  - Main 1
  - Movie
  - MovieService

Seat.class | BookingService.java

BookingService.java > ...

```java
public class BookingService {
    public void bookSeat(Show show, int r, int c) {
        Seat seat = show.getSeat(r, c);
        if (seat == null) {
            System.out.println(x: "Invalid seat");
            return;
        }
        seat.book();
    }
}
```

PROBLEMS 1 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\ASUS\OneDrive\Documents\MovieBookingSystem> javac *.java
>> java Main
1. View Movies
2. Book Ticket
3. Exit
1
1. Inception
2. Interstellar
1. View Movies
2. Book Ticket
3. Exit
2
1. Inception
2. Interstellar
Select movie:
2
A A A A A
A A A A A
A A A A A
A A A A A
A A A A A
Enter row and column:
2 2
Seat booked
1. View Movies
2. Book Ticket
3. Exit
```

Thank You